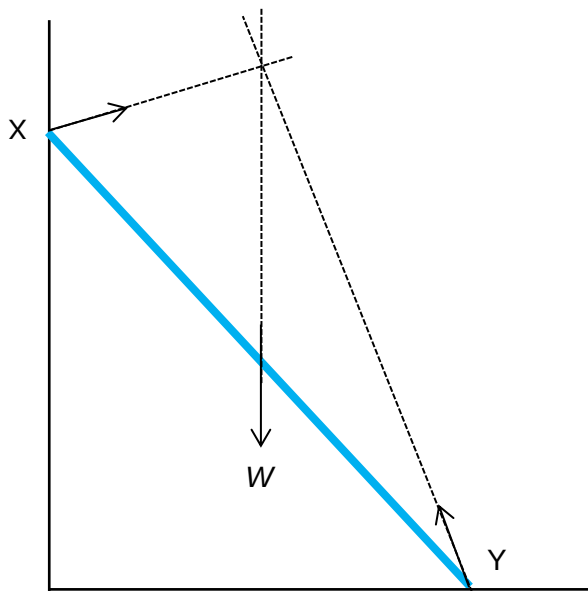


Q3	
(a)	Correct labelling of forces As speed of metal ball increases, viscous force increases. As viscous force increases, net acceleration decreases. When the total upward force (viscous force and upthrust) becomes equal in magnitude to the weight of the ball, there is no net force, and the metal ball reaches terminal velocity.
(b)(i)	Taking moments about B, $P \times 6.0 \cos 30^\circ = 150 \times 3.0 \sin 30^\circ$ $P = 43.301 \text{ N}$ = 43 N (2 s.f.) shown
(b)(ii)	The horizontal component of Q which is pointing to the left will balance force P which is pointing to the right. The vertical component of Q which is acting upwards will balance W which is acting downwards vertically. This will allow the ladder to be in equilibrium.
(b)(iii)	Horizontal component = 43 N Vertical component = 150 N Magnitude of $Q = \sqrt{43^2 + 150^2} = 156 \text{ N (3 s.f.)}$

(b)(iv)



Three forces intersect at same point and force at X is facing upwards diagonally.